

DBbun Invisible Illness Burden Simulator: quality assurance log

Uri Kartoun, DBbun LLC. Covers work through 10 October 2026. Independent entry, not endorsed by HHS or NIH.

How quality was checked

Check	Who or what did it	What it found	What changed
Automated tests	Python test suite (unittest)	220 tests pass, 5 skipped.	Every defect fixed gets a regression test.
Own five-part audit	Claude Code with Claude Sonnet 5.5; read-only first, reports kept	Accuracy, function, federal-resource, interface and wording problems.	Fixed in batches, then re-tested.
Independent model critiques of every simulator PDF (25) and comparison PDF (13)	Claude Sonnet 5.5, Grok 4.5, Gemini 3.5 (Google), GPT (OpenAI; exact build not exposed) and GPT-5 Codex, given one rubric-based prompt	Reviewers rated the comparison PDFs 3 to 4 of 5 and raised the same themes across simulators: price-year and currency labels, non-U.S. data compared with U.S. figures, text that broke the PDF layout, over-confident model wording.	Fixed once in the code and prompts, not simulator by simulator. All 25 simulators were then regenerated and the 13 comparisons rebuilt (about \$21 in model calls).
Independent code review	GPT-5 Codex; report only, it changed no code	Ten findings.	Nine fixed with regression tests; one documented (the browser token is a single-user limit).
Federal cross-checks	Code, not the AI model	Each figure a paper states is restated in current prices or compared with CDC and survey data.	Mismatches are shown on page 1 of each PDF, not hidden.
FHIR export	Official HL7 FHIR validator, R4 4.0.1, terminology checking off	26 files (25 simulators and the third-condition demo): 0 errors. 40 to 74 best-practice warnings each: no patient, no performer or time, no narrative text, DBbun-defined codes.	First run gave 28 errors per file (undefined DBbun extensions); fixed by shipping their definitions with each bundle.
Public demo	Rebuilt from the same code and opened live	25 simulators, 13 comparisons and the third-condition demo load; FAQ and downloads match the local tool.	Rebuilt and published 10 October 2026, with the audio overviews, simulator numbering and the feedback log.

How reviewer advice was handled

I fixed only problems that were real and systematic, and checked each against the paper or the data first. Suggested wording that claimed more than the evidence showed (stronger causal or federal-comparison language, or implied certification) was not used. The reviewers are tools, not authorities: I decided what to adopt.

What this log does not show

- The outside reviews covered the earlier versions. The regenerated simulators were checked by the tests, the cross-checks and my own spot checks, but have not had a second full outside review.
- No independent person has yet completed a documented, uncoached task in the public demo. Two outside readers, a member of a neighborhood group and a person with lived experience of Lyme disease and a research background, gave written feedback without doing the tasks: the first screen was too dense, the page was text-heavy, and my time estimate was too short. In response I added a note telling new visitors to skip the grid, collapsed the coverage map, added a one-line summary under each study, a Lyme / Long COVID filter, a "what this is and is not" banner and FAQ pictures, and shortened my request to one simulator. I am still asking in online communities; further results will be added here only if they happen.
- The 38 audio overviews (one per simulator and one per comparison) are read by a synthetic voice from scripts composed from each saved simulator or comparison record. Each script has a source table mapping its paragraphs to the document or saved record; the first three scripts were also reviewed paragraph by paragraph against their documents with Claude Code, and the rest by automated rules only. The FHIR export is a prototype. It is not clinical data, not profiled to US Core or USCDI+, and not certified. Terminology checking was off, so DBbun-defined codes were not checked against a terminology server.
- The two demonstrations on other conditions (a paper on post-surgical chronic pain and an emergency-care cost paper) used the general pipeline only. The cost paper's dollar amounts were restated in current prices by the price-index check; neither has curated federal evidence for its condition or outside validation.

AI tools and model IDs

Inside the product: Claude Sonnet 5 (claude-sonnet-5, Anthropic) reads the papers and writes each specification and its wording; code validates it and does all the arithmetic. Building and testing: Claude Code (Anthropic) with Claude Sonnet 5.5 (claude-sonnet-5-5). Audio overviews: scripts written with Claude Code; voice by Kokoro-82M, an open-source text-to-speech model run locally. Reviews: Claude Sonnet 5.5, Grok 4.5 (xAI), Gemini 3.5 (Google), GPT and GPT-5 Codex (OpenAI; exact builds not exposed). Interviewees, forum respondents and outside testers are described by role only.